Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 16)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 1 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 1 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 1 . 3 Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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0 1 . 4 Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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0 2 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking .....

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DNA databases and eDNA .....

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Image recognition software .....

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Acoustic monitoring / sonograms .....

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0 2 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 3 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed .....

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Sea lice and disease .....

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Escapes affecting wild gene pools .....

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Organic waste pollution .....

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0 3 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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0 4 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 4 . 1 Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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0 4 . 2 Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

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0 4 . 3 Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

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0 4 . 4 Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

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0 5 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 5 . 1 Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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0 5 . 2 Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 5 . 3 Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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0 6 A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 6 . 1 Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

0 6 . 2 Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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0 6 . 3 Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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0 7 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 7 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

0 7 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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0 8 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 8 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 8 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 9 . 1 Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

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0 9 . 2 Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

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0 9 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores .....

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Gas analysis of ice cores .....

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Radiocarbon dating .....

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Magnesium : calcium ratio in calcite .....

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1 0 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

1 0 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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1 0 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

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1 0 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

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1 0 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS